# Supplementary Material

## TARVI: Transcription-Factor Aided RNA Velocity Inference with Supervised Latent Time and Velocity-Pseudotime Blending

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This supplement gives the full component ablation that the main paper summarises. We first set out the background theory the analysis builds on—the closed-form kinetics and the velocity-graph pseudotime that TARVI uses—at a level of detail condensed in the main text, then present the ablation and the reproducibility and scope notes that follow from it. Section and table numbering carry an **S** prefix and are local to this document.

## S1. Background Theory

**Closed-form four-state kinetics.** TARVI inherits VeloVI's two-branch solution of the splicing ODEs, which the main text states only in words. With transcription, splicing and degradation rates $\alpha, \beta, \gamma$ and a per-gene switching time $t_s$, the induction branch starts from $u{=}s{=}0$ and evolves over $t_{\text{ind}} = t_s \rho_{i,g}$ as

$$u_{\text{ind}}(t) = \tfrac{\alpha}{\beta}\big(1 - e^{-\beta t}\big), \qquad s_{\text{ind}}(t) = \tfrac{\alpha}{\gamma}\big(1 - e^{-\gamma t}\big) + \tfrac{\alpha}{\gamma-\beta}\big(e^{-\gamma t} - e^{-\beta t}\big), \tag{1}$$

and the repression branch starts from the induction endpoint $(u_0, s_0) = (u_{\text{ind}}(t_s), s_{\text{ind}}(t_s))$ and evolves over $t_{\text{rep}} = (t_{\max} - t_s)\tau_{i,g}$ as

$$u_{\text{rep}}(t) = u_0\, e^{-\beta t}, \qquad s_{\text{rep}}(t) = s_0\, e^{-\gamma t} - \tfrac{\beta u_0}{\gamma-\beta}\big(e^{-\gamma t} - e^{-\beta t}\big), \tag{2}$$

(a small $\varepsilon$ regularises $\gamma - \beta$), with the steady-state branches as in the main paper. The RNA velocity is $v = \beta u - \gamma s$ evaluated on these curves; in TARVI $\alpha$ is the cell-specific, TF-regulated rate $\alpha_i$, so Eqs. (1)–(2) are evaluated per cell. These curves are what the Gene-$\mathrm{R}^2_{\text{spl}}$ metric scores, which is why the TF rate enters the kinetic-fit discussion below.

**Velocity-graph pseudotime.** The blending step builds its global coordinate not from expression distances but from the *velocity* transition graph. For each cell $i$ and neighbour $j$ in its $k$NN set $\mathcal{N}(i)$, a directed transition probability comes from the cosine alignment of the velocity $\mathbf{v}_i$ with the displacement $\mathbf{s}_j - \mathbf{s}_i$,

$$\pi_{ij} \;=\; \frac{\exp\big(\cos(\mathbf{v}_i,\ \mathbf{s}_j - \mathbf{s}_i)/\sigma\big)}{\sum_{k\in\mathcal{N}(i)} \exp\big(\cos(\mathbf{v}_i,\ \mathbf{s}_k - \mathbf{s}_i)/\sigma\big)}, \qquad \sigma > 0, \tag{3}$$

giving a row-stochastic matrix that points downhill along the field [1]; $t^{\text{VPT}}$ is then the diffusion pseudotime [2] from a root cell, rescaled to $[0, 1]$. Because $\Pi = [\pi_{ij}]$ is built from the *already-trained* velocities, the blend is a post-hoc step that leaves the velocity field unchanged—the property the ablation exploits.

## S2. Component Ablation

We ablate each of the three components—the TF-regulated rate (TF), the velocity–time consistency loss (VTC), and the velocity-pseudotime blend (VPT)—across all five datasets (pancreas, bone marrow, forebrain, chromaffin, scEU-organoid). Table S1 reports cross-dataset means for the VeloVI baseline; VeloVI with the blend on its *own* velocity graph (VeloVI + VPT); the full model with one component removed; and full TARVI. We write $\Delta_c\, m$ for the change in a metric $m$ when component $c$ is removed from the full model, and $\delta_{\mathrm{VPT}}\, m$ for the change from adding the blend to the raw VeloVI graph.

Table S1: Component ablation: cross-dataset mean performance over all five datasets. VeloVI + VPT applies the blend ($\beta$=0.7) to the unmodified VeloVI velocity graph. Best per column among configurations in bold. All metrics ↑.

| Config | ICCoH | CBDir | VelConf | PT-Spear | PT-DistCorr | PT-Cons | RootAcc | Gene-$R^2_{\mathrm{spl}}$ |
|---|---|---|---|---|---|---|---|---|
| VeloVI | 0.787 | 0.931 | 0.920 | 0.331 | 0.341 | 0.905 | 0.723 | 0.442 |
| VeloVI + VPT | 0.787 | 0.931 | 0.920 | 0.714 | 0.724 | **0.986** | 0.658 | 0.442 |
| TARVI − TF | 0.693 | 0.911 | 0.902 | **0.759** | **0.783** | 0.984 | 0.653 | 0.461 |
| TARVI − VTC | **0.792** | **0.934** | **0.928** | 0.700 | 0.705 | **0.986** | 0.556 | 0.474 |
| TARVI − VPT | 0.776 | 0.933 | 0.927 | 0.320 | 0.316 | 0.891 | **0.728** | **0.479** |
| TARVI (full) | 0.776 | 0.933 | 0.927 | 0.747 | 0.772 | **0.986** | 0.671 | **0.479** |

The three components act on different axes. **The blend supplies the pseudotime ordering:** $\Delta_{\mathrm{VPT}}$(PT-Spear) = 0.747 − 0.320 = 0.427, with every velocity-field metric (ICCoH, CBDir, VelConf, Gene-$R^2_{\mathrm{spl}}$) unchanged between full and −VPT—exactly as the post-hoc construction of Eq. (3) predicts. The effect is largely model-agnostic: on the raw VeloVI graph it already gives $\delta_{\mathrm{VPT}}$(PT-Spear) = 0.714 − 0.331 = 0.383, and the gap between VeloVI + VPT and full TARVI (+0.033) is the part specific to TARVI's velocity field. We therefore treat the blend as an orthogonal enhancement that recovers a well-ordered pseudotime from any reasonable velocity graph.

**The TF rate sharpens the velocity field** rather than the temporal coordinate: $\Delta_{\mathrm{TF}}$ is positive on ICCoH (+0.083), VelConf (+0.025) and CBDir (+0.022) but slightly negative on PT-Spear. Its kinetic-fit gain on the curves of Eqs. (1)–(2) is real but modest ($\Delta_{\mathrm{TF}}$(Gene-$R^2_{\mathrm{spl}}$) = 0.018), by design: the basal rate is warm-started from VeloVI and frozen so the TF term is an incremental correction, VeloVI's four-state mixture already fits per-gene spliced dynamics well, and the L1 penalty keeps the weights sparse. The cell-specific rate thus behaves as a biologically grounded regulariser that improves field consistency more than it inflates $R^2$. **The VTC loss improves temporal ordering and root/terminal identification:** $\Delta_{\mathrm{VTC}}$(PT-Spear) = 0.047 and, most clearly, $\Delta_{\mathrm{VTC}}$(RootAcc) = 0.115, with a negligible effect on the field metrics ($\leq 0.016$).

Finally, the blend weight $\beta$ is robust. The two endpoints of the convex blend appear directly in the table—$\beta$=0 (time-head only) is TARVI − VPT (PT-Spear 0.320) and $\beta$=1 (pure velocity-graph pseudotime) is VeloVI + VPT (0.714)—and the global value $\beta$=0.7 used throughout (full TARVI, 0.747) exceeds both. The learned time and the velocity-graph pseudotime are therefore complementary, and $\beta$=0.7 is a safe interior operating point chosen once on the pooled cross-dataset mean rather than tuned per dataset.

## S3. Qualitative Comparison on Chromaffin and Forebrain

Complementing the pancreas case study in the main paper, Figs. S1 and S2 show, for four methods, the projected velocity streamlines (top row) and the inferred latent time (bottom row) on the chromaffin (t-SNE) and forebrain (UMAP) datasets; all panels are rendered identically (shared cluster palette) and titled by method. On both datasets TARVI produces coherent streamlines together with a smooth $0 \rightarrow 1$ latent-time gradient, consistent with its temporal-metric lead (chromaffin PT-Spear 0.807, forebrain 0.887, versus 0.147 and 0.342 for VeloVI). VeloVI recovers a similar field (chromaffin/forebrain CBDir 0.911/0.938) but an uncalibrated latent time; scVelo dynamical and Velocyto give intermediate results. TFvelo is omitted from these panels as it does not export per-cell velocities in a re-loadable form; its quantitative scores are in the main paper.

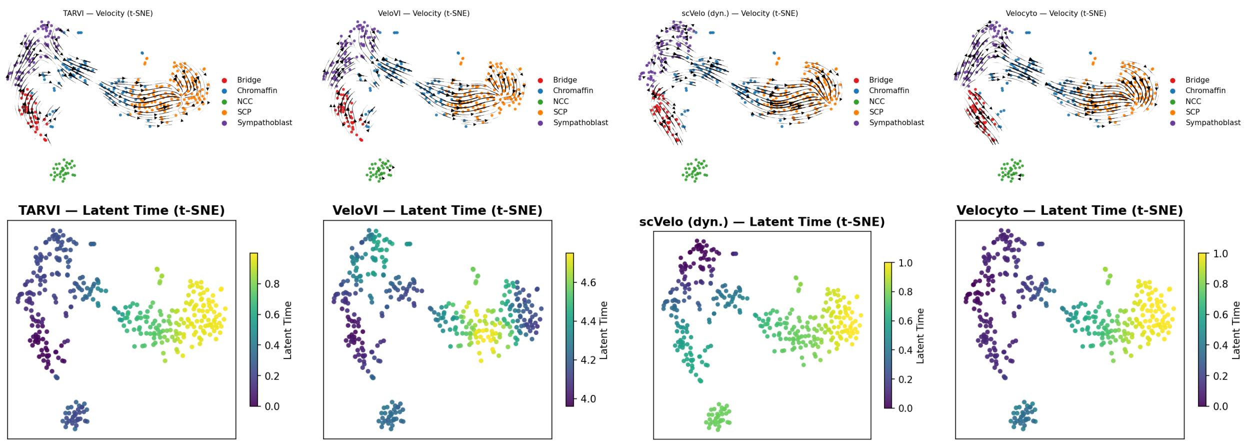


Figure S1: Chromaffin (t-SNE): velocity streamlines (top) and inferred latent time (bottom) for each method; panels are titled by method.

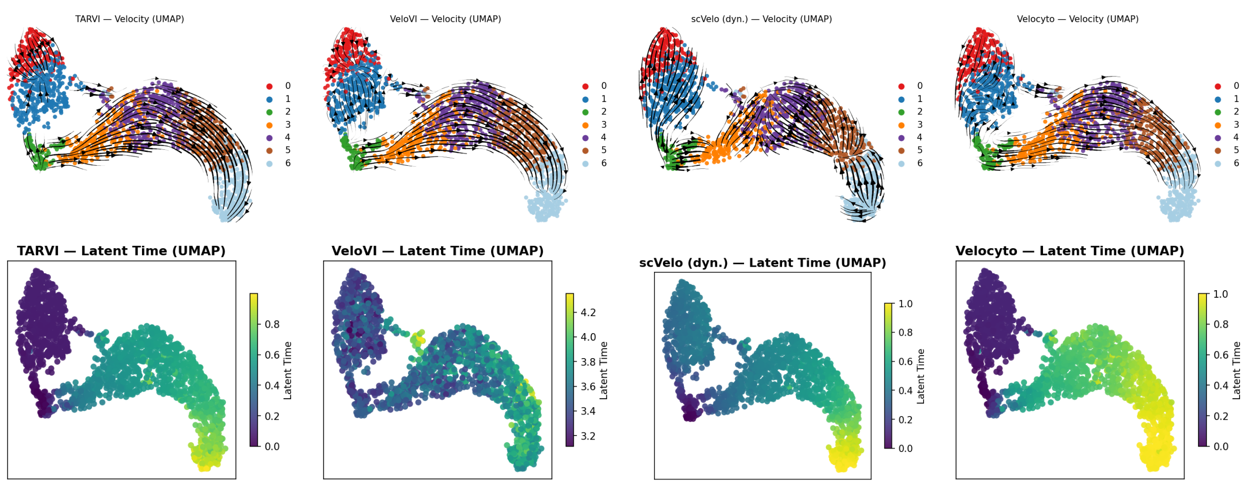


Figure S2: Forebrain (UMAP): velocity streamlines (top) and inferred latent time (bottom) for each method; panels are titled by method.

## S4. Reproducibility, Statistical Reading, and Scope

All results use a fixed random seed and a deterministic preprocessing pipeline, and we report single-run point estimates. The values in Table S1 come from a dedicated ablation run and differ from the main-text headline numbers by $\leq 0.01$ on the aggregate means; comparisons within the table are internally consistent. This $\leq 0.01$ drift is the right scale for reading small differences:

gaps on that order—such as TARVI versus VeloVI on the directional metrics (CBDir, VelConf)—should be read as comparable rather than as wins, whereas the pseudotime contributions above ($\Delta_{\mathrm{VPT}}$ PT-Spear $= 0.427$) are an order of magnitude larger.

A few limitations bound the scope of these conclusions. We did not run a multi-seed study, so we report no confidence intervals or paired significance tests; a multi-seed analysis with paired tests across datasets is the natural next step [3]. The TF prior is rebuilt per dataset by intersecting the ENCODE/ChEA databases with each gene panel; because such databases are incomplete and tissue-context dependent, genes with no reliable TF edges fall back (through the frozen basal rate and the L1 penalty) to VeloVI-like behaviour, so for lineages or organisms outside the ENCODE/ChEA scope we expect TARVI to approach the no-TF regime rather than to fail, and a systematic study of mask completeness is left to future work. Finally, VeloTrace [4], a concurrent method targeting the same velocity–trajectory gap, has no public code release that would allow a faithful re-implementation under our preprocessing and metric protocol, so we could not include it among the baseline comparisons.

## References


[1] V. Bergen *et al.*, "Generalizing RNA velocity to transient cell states through dynamical modeling," *Nature Biotechnology*, vol. 38, pp. 1408–1414, 2020.

[2] L. Haghverdi *et al.*, "Diffusion pseudotime robustly reconstructs lineage branching," *Nature Methods*, vol. 13, pp. 845–848, 2016.

[3] J. Liu *et al.*, "Comprehensive benchmarking of RNA velocity methods across single-cell datasets," 2026.

[4] H. Cheng *et al.*, "VeloTrace Reconciles Divergent Velocity and Trajectory in Single-cell Transcriptomics with Deep Neural ODE," *bioRxiv*, 2026.